Lecture 7 9/15/22 more contradiction

Want to prove P.

Assume $\neg P$.

Argue.

Conclude Q.

$\neg P \Rightarrow Q$

Observe that Q is false

Conclude P.

Prove that $(\forall n \in \mathbb{Z}$, n and $n+1$ have no common prime factors$)$

<u>Proof</u>: Proceed by contradiction. Assume $\exists n \in \mathbb{Z}$ s.t. n and $n+1$ have some common prime divisors. Let p be a prime s.t. $\underline{p \mid n}$ and $\underline{p \mid n+1}$.

Then $p \mid (n+1) - n$, i.e., $p \mid 1$.
This is a contradiction, since p is prime, and primes are >1. $\blacksquare$

$\Rightarrow \Leftarrow$ ↯ ⨳

$d \mid a \wedge d \mid b \Rightarrow d \mid a+b$
$d \mid a \Rightarrow d \mid -a$
$d \mid ab$ $\quad d \mid a-b$

Let $a, b, c \in \mathbb{Z}$. Sps $a^2 + b^2 = c^2$. Hyp

Show that abc is even.

Proof: Proceed by contradiction.

Assume $a^2+b^2=c^2$ and abc is odd.

Then a, b, and c are each odd. (BK if one were even, abc would be even)

Then a^2, b^2, & c^2 are odd (b/c products of odd integers are odd). Then a^2+b^2 is even, but c^2 is odd. This is "even = odd" which is a contradiction. $\blacksquare$

$\boxed{\text{\underline{Defn}: A number } x \text{ is \underline{rational} if } \exists\, a,b \in \mathbb{Z} \text{ s.t. } b \neq 0 \text{ and } x = a/b.}$

A rational # is <u>reduced</u> if a and b have no common prime divisors

<u>Examples</u>: $\frac{2}{3} = \frac{4}{6}$

reduced   not reduced.

<u>Fact</u>: every rational # can be reduced

Prove: $\sqrt{2} \notin \mathbb{Q}$. (It $\sqrt{2}$ is not Rational.)

Proof: Proceed by contradiction. Assume $\sqrt{2} \in \mathbb{Q}$.
Then $\exists\, a, b \in \mathbb{Z}$ s.t. $b \neq 0$ and $\sqrt{2} = a/b$.
Assume that a & b are reduced. In particular at least one of a or b is odd. Then

$b\sqrt{2} = a$. Then $b^2 \cdot 2 = a^2$.

Since the LHS is even, the RHS is even, i.e.,
a^2 is even. Thus a is even. (By HW1, if
a is odd a^2 is odd.) Then $4 \mid a^2$; indeed,
if $d \mid e$ + $f \mid g$ then $df \mid eg$). Write $a = 2k$

$b^2 \cdot 2 = a^2$

for some $k \in \mathbb{Z}$. Then $b^2 \cdot 2 = (2k)^2 = 4k^2$.
Then $b^2 = 2k^2$. Since the RHS is even, b^2 is even,
So b is even. This is a contradiction,

Since $a+b$ are not both odd.

HW: $\sqrt{3} \notin \mathbb{Q}$

$3^{2/3}$

$\tan 1$

$\pi = 3.141\ldots$

e

Prove: $\log_3 2 \notin \mathbb{Q}$.

Recall: $3^{\log_3 2} = 2$

Proof: Proceed by contradiction.

Assume $\log_3 2 \in \mathbb{Q}$. Then $\exists\, a, b \in \mathbb{Z}$ s.t. $b \neq 0$ and $\log_3 2 = \frac{a}{b}$. WMA (we may assume) a & b are reduced.

Then $2 = 3^{\log_3 2} = 3^{a/b}$.

Then $2^b = 3^a$. $\left((3^{a/b})^b = 3^a \right)$

Since $b > 0$, the LHS is even.

But the RHS is odd. This is

a contradiction. $\blacksquare$

Prove: if $a \in \mathbb{Q}$ and $b \notin \mathbb{Q}$, then $a+b \notin \mathbb{Q}$.

Proof: Assume $a \in \mathbb{Q}$ and $b \notin \mathbb{Q}$. Proceed by contradiction. Assume $a+b \in \mathbb{Q}$. Then $\exists\, c,d \in \mathbb{Z}$ s.t. $d \neq 0$ and $a = c/d$. Then $\exists\, e,f \in \mathbb{Z}$ s.t. $f \neq 0$ and $a+b = e/f$

Then $b = (a+b) - a = \frac{e}{f} - \frac{c}{d} = \frac{ed - cf}{fd}$.

Since $ed - cf, fd \in \mathbb{Z}$ and $fd \neq 0$,

$b \in \mathbb{Q}$. This contradicts $b \notin \mathbb{Q}$. $\blacksquare$

We conclude that $a+b \notin \mathbb{Q}$.

Let a, b, c be odd. Let x be a solution to $ax^2 + bx + c = 0$. Prove $x \notin \mathbb{Q}$.

<u>Proof:</u> Assume a, b, c are odd.

Assume $ax^2 + bx + c = 0$.

Proceed by contradiction. Assume $x \in \mathbb{Q}$.

Then $\exists d, e \in \mathbb{Z}$ s.t. $e \neq 0$ and $x = d/e$. WMA d & e are reduced.

Then $a\left(\frac{d}{e}\right)^2 + b\frac{d}{e} + c = 0$. \begin{equation}

Then $ad^2 + bde + ce^2 = 0$. (1.1)

Since d & e are reduced, at least one is odd.

There are 3 cases: d & e are odd, d is even & e is odd, or d is odd & e is even.

In case 1, the LHS of (1.1) is "odd + odd + odd = odd ≠ even", a contradiction.

In case 2, the LHS is "e + e + o = e", ↯

Case 3 is similar.

In each case, we have a contradiction ∎

2, 3, 5, 7, 11 ...

p is <u>prime</u> if "$p = ab \Longrightarrow a = \pm 1$ or $b = \pm 1$"

$p \neq 1$ and $\forall a, b$ s.t. $p = ab$, $a = \pm 1$ or $b = \pm 1$

<u>~~Fermat~~</u>:

$1 = 11 = 111 = \ldots$

Fermat:

$$2^0+1 = 1+1 = 2$$

$$2^1+1 = 2+1 = 3$$

$$2^2+1 = 4+1 = 5$$

$$2^4+1 = 16+1 = 17$$

$$2^8+1 = 256+1 = 257$$

$$2^{16}+1 = 65536+1 = 65537$$

Fermat conj'd that $\forall n$, $2^{2^n}+1$ is prime.

False

Modern conj: $2^{2^n}+1$ is never prime if $n \geq 5$.

$$2^3+1 = 8+1 = 9 = 3\cdot 3$$

$$2^5+1 = 32+1 = 33 = 3\cdot 11$$

$$2^7+1 = 128+1 = \underline{\underline{129}} = 3\cdot 43$$

Problem: if 2^n+1 is prime then n is even.

Proof: Assume 2^n+1^n is prime.

Proceed by contradiction. Assume n is odd.

$\Bigg(n^2-1 \to (n-1)(n+1)$

$x^n - y^n = (x-y)(x^{n-1} + x^{n-2}y + \cdots y^{n-1}) \Bigg)$

Since n is odd, $(-1)^n = -1$.

Thus $2^n + 1 = 2^n + 1^n = 2^n - (-1)^n$.

This factors as $\big(2-(-1)\big)\left(2^{n-1} - 2^{n-2} + \cdots \pm 1\right)$

$$= 3 \cdot ?$$

If $2^n + 1 > 3$, it is not prime b/c 3 is a proper divisor. $\boxed{}$